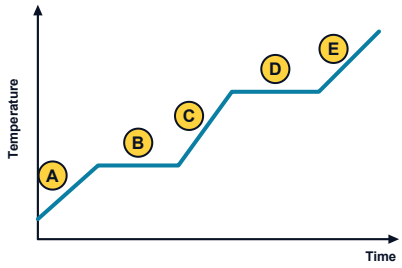
DO NOW · AO1 REVISION

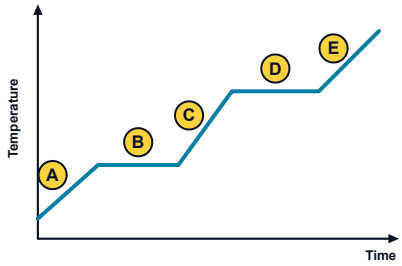
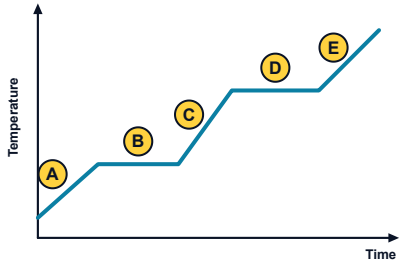
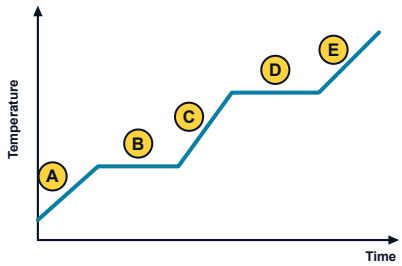
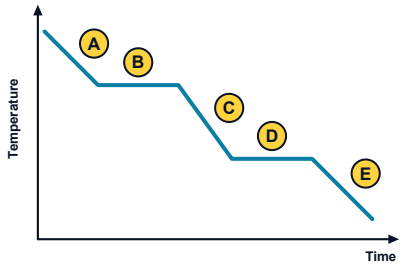
Changes of state and specific latent heat

Particle model of matter
Paper 1 · Combined and Separate
14 questions

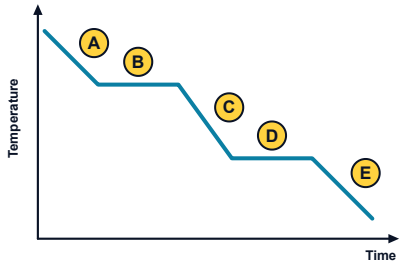
Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 4 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	**State the standard unit of specific latent heat.**	Joule per kilogram (J/kg)	☐
2 ★	**Define latent heat.**	The energy needed to change the state of a substance with no change in temperature.	☐
3 ★	**Define specific latent heat.**	The energy needed to change the state of 1 kg of a substance with no change in temperature.	☐
4	**Define specific latent heat of fusion.**	The energy needed to change 1 kg of a substance from solid to liquid with no change in temperature.	☐
5	**Define specific latent heat of vaporisation.**	The energy needed to change 1 kg of a substance from liquid to vapour (gas) with no change in temperature.	☐
6	**Which letter on the heating curve shows the substance melting?** 	B	☐

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	QUESTION	ANSWER	✓
7	**Which letter on the heating curve shows the substance boiling?** 	D	☐
8	**Which letter on the heating curve shows a solid getting hotter?** 	A	☐
9	**Which letter on the heating curve shows a liquid getting hotter?** 	C	☐
10	**Which letter on the cooling curve shows the substance freezing?** 	D	☐

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	QUESTION	ANSWER	✓
11	**Which letter on the cooling curve shows a liquid cooling?** 	C	☐
12	**State the equation linking energy for a change of state, mass and specific latent heat.**	energy for a change of state = mass × specific latent heat (E = m L)	☐
13 ★	**Explain why the temperature stays constant while a substance melts.**	The energy supplied increases the potential energy of the particles, not their kinetic energy.	☐
14	**State the difference between specific heat capacity and specific latent heat.**	Specific heat capacity: energy to change the temperature of 1 kg by 1 °C. Specific latent heat: energy to change the state of 1 kg.	☐

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